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United States Patent	12409707
Kind Code	B2
Date of Patent	September 09, 2025
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Vehicular air conditioning system

Abstract

A vehicular air conditioning system includes an intake case having an indoor air inlet for introducing an indoor air present in a vehicle interior and an outdoor air inlet for introducing an outdoor air present outside a vehicle, an intake door installed in the intake case to selectively block one of the outdoor air inlet and the indoor air inlet while rotating about a rotation center shaft, and a water blocking part configured to block water flowing into the vehicle interior from the indoor air inlet of the intake case.

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Appl. No.:	17/921497
Filed (or PCT Filed):	June 24, 2021
PCT No.:	PCT/KR2021/007950
PCT Pub. No.:	WO2022/005103
PCT Pub. Date:	January 06, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230158862 A1	May. 25, 2023

Foreign Application Priority Data

Publication Classification

Int. Cl.: B60H1/26 (20060101)

U.S. Cl.:

CPC B60H1/26 (20130101);

Field of Classification Search

CPC: B60H (2001/00085); B60H (1/00849); B60H (1/26); B60H (1/00685); B60H (2001/00707); B60H (2001/00721); B60H (2001/00714); B60H (1/3233)

USPC: 454/143

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a national phase under 35 U.S.C. § 371 of International Application No. PCT/KR2021/007950 filed on Jun. 24, 2021, which claims the benefit of priority from Korean

Patent Application Nos. 10-2020-0079764 filed on Jun. 30, 2020 and 10-2020-0079747 filed on Jun. 30, 2020. The entire contents of these applications are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(2) The present invention relates to a vehicular air conditioning system, and more particularly, a vehicular air conditioning system in which the movement of water from an outdoor air inlet to an indoor air inlet can be limited to reliably prevent the flow of water from the outdoor air inlet into the indoor air inlet and consequently the flow of water from the outdoor air inlet into the vehicle interior.

BACKGROUND ART

(3) As shown in FIG. 1, a motor vehicle includes an air conditioning system for cooling and heating a vehicle interior. The air conditioning system includes an intake part **10** that sucks an indoor air or an outdoor air and blows it into the vehicle interior.

(4) The intake part **10** is usually installed on the engine room side with respect to a dash panel D. The intake part **10** includes an intake case **12**. The intake case **12** is provided with an indoor air inlet **12a** and an outdoor air inlet **12b**. An intake door **14** is installed between the indoor air inlet **12a** and the outdoor air inlet **12b**.

(5) The indoor air inlet **12a** communicates with the vehicle interior to introduce indoor air, and the outdoor air inlet **12b** communicates with the vehicle exterior to introduce outdoor air.

(6) Typically, the indoor air inlet **12a** is formed on the side surface portion of the intake case **12** corresponding to the vehicle interior, and the outdoor air inlet **12b** is formed on the upper surface portion of the intake case **12** corresponding to the vehicle exterior. Therefore, the outdoor air inlet **12b** is located more upward than the indoor air inlet **12a** in the gravity direction.

(7) The intake door **14** is a dome-shaped door that rotates between the outdoor air inlet **12b** and the indoor air inlet **12a** to selectively open the outdoor air inlet **12b** or the indoor air inlet **12a**.

(8) In particular, the intake door **14** blocks the outdoor air inlet **12b** as it moves toward the upper surface portion of the intake case **12**, and blocks the indoor air inlet **12a** as it moves toward the side surface portion of the intake case **12**. Therefore, the indoor air inlet **12a** and the outdoor air inlet **12b** can be selectively opened to selectively introduce the outdoor air or the indoor air.

(9) In addition, the intake part **10** includes an air filter **20** and a blower **22**. The air filter **20** filters foreign substances in the indoor air or the outdoor air introduced through the indoor air inlet **12a** or the outdoor air inlet **12b**. The blower **22** sucks the indoor air or the outdoor air through the indoor air inlet **12a** or the outdoor air inlet **12b**, and blows the sucked indoor air or outdoor air toward the vehicle interior.

(10) However, in such a conventional air conditioning system, it is sometimes the case that external water W, for example, rainwater, is introduced to the outdoor air inlet **12b**. The water thus introduced may flow into the vehicle interior through the intake door **14** and the inner wall surface of the intake case **12**.

(11) In particular, when the intake door **14** is blocking the outdoor air inlet **12b**, the water W introduced into the outdoor air inlet **12b** remains on the upper surface of the intake door **14** as it is. In this state, if the intake door **14** is rotated toward the indoor air inlet **12a** on the side surface portion of the intake case **12**, the water accumulated on the upper surface of the intake door **14** flows toward the indoor air inlet **12a** and then flows into the vehicle interior.

(12) In addition, around the edge of the intake door **14**, as shown in FIGS. 1 and 2, a seal **14a** for sealing a gap between the intake door **14** and the intake case **12** is installed. Since the seal **14a** is in close contact with the inner wall of the intake case **10**, the water W introduced into the outdoor air inlet **12b** is not drained into the interior of the intake case **12** and is accumulated on the side surface portion of the intake door **14**.

(13) The water W accumulated in this way is infiltrated into the gaps between the seals **14a** on both sides of the intake door **14** and the door sealing surfaces **12c** and **12d** of the intake case **12**

corresponding thereto, and is moved toward the indoor air inlet **12a** located on the lower side in the direction of gravity. The water W thus moved flows into the vehicle interior through the indoor air inlet **12a**.

SUMMARY

(14) In view of the problems inherent in the related art, it is an object of the present invention to provide a vehicular air conditioning system capable of improving the internal structure to prevent the movement of water from an outdoor air inlet to an indoor air inlet.

(15) Another object of the present invention is to provide a vehicular air conditioning system capable of limiting the movement of water from the outdoor air inlet toward the indoor air inlet when the water is introduced into the outdoor air inlet.

(16) A further object of the present invention is to provide a vehicular air conditioning system capable of preventing the water from flowing from the outdoor air inlet into the vehicle interior through the indoor air inlet.

(17) A still further object of the present invention is to provide a vehicular air conditioning system capable of rapidly discharging the water introduced onto the upper surface of an intake door.

(18) A yet still further object of the present invention is to provide a vehicular air conditioning system capable of preventing the water from being accumulated on the upper surface of an intake door.

(19) An even yet still further object of the present invention is to provide a vehicular air conditioning system capable of preventing the water accumulated on the upper surface of an intake door from flowing into the vehicle interior.

(20) In order to achieve these objects, there is provided a vehicular air conditioning system, including: an intake case having an indoor air inlet for introducing an indoor air present in a vehicle interior and an outdoor air inlet for introducing an outdoor air present outside a vehicle; an intake door installed in the intake case to selectively block one of the outdoor air inlet and the indoor air inlet while rotating about a rotation center shaft; and a water blocking part configured to block water flowing into the vehicle interior from the indoor air inlet of the intake case.

(21) In the system, the outdoor air inlet may be located more upward than the indoor air inlet in a gravity direction, and the water blocking part may be configured to prevent the water introduced into the outdoor air inlet from flowing into the vehicle interior while moving toward the indoor air inlet.

(22) In the system, the water blocking part may include a water storage groove formed in a portion of the intake case below the indoor air inlet in a water passage between the outdoor air inlet and the indoor air inlet, and configured to store the water moving from the outdoor air inlet toward the indoor air inlet.

(23) In the system, the water storage groove may be configured to store the water that flows toward a bottom surface of the indoor air inlet when the water accumulated on the upper surface of the intake door flows to the bottom surface of the indoor air inlet during the rotational movement of the intake door.

(24) In the system, an outdoor air inlet side door sealing surface and an indoor air inlet side door sealing surface on which the intake door is seated and sealed may be formed on the inner wall surfaces of the intake case, and the water storage groove may be configured to store the water that flows from the outdoor air inlet toward the indoor air inlet along the outdoor air inlet side door sealing surface and the indoor air inlet side door sealing surface of the intake case.

(25) The system may further include: a drain passage configured to drain the water collected in the water storage groove. The drain passage may include a drain hole formed to extend from a bottom surface of the water storage groove toward the inner lower side of the intake case to drain the water stored in the water storage groove to the inner lower side of the intake case.

(26) According to the vehicular air conditioning system of the present invention, the water moving from the outdoor air inlet to the indoor air inlet is stored and then quickly drained. Therefore, it is

possible to limit the movement of water from the outdoor air inlet to the indoor air inlet.
(27) In addition, since the movement of water from the outdoor air inlet to the indoor air inlet can be limited, it is possible to reliably prevent the flow of water from the outdoor air inlet into the indoor air inlet and consequently the flow of water from the outdoor air inlet into the vehicle interior.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a side sectional view showing a conventional vehicular air conditioning system.
- (2) FIG. 2 is a sectional view taken along line II-II in FIG. 1, showing the conventional vehicular air conditioning system.
- (3) FIG. 3 is a side sectional view showing a vehicular air conditioning system according to a first embodiment of the present invention.
- (4) FIG. 4 is a sectional view taken along line IV-IV in FIG. 3, showing the vehicular air conditioning system according to the first embodiment of the present invention.
- (5) FIG. 5 is a perspective view showing main features of the vehicular air conditioning system according to the first embodiment of the present invention.
- (6) FIG. 6 is a side sectional view showing a vehicular air conditioning system according to a second embodiment of the present invention.
- (7) FIG. 7 is a sectional view taken along line VII-VII in FIG. 6, showing the vehicular air conditioning system according to the second embodiment of the present invention.
- (8) FIG. 8 is a perspective view showing the main features of the vehicular air conditioning system according to the second embodiment of the present invention.

DETAILED DESCRIPTION

(9) Preferred embodiments of a vehicular heat management system according to the present invention will now be described in detail with reference to the accompanying drawings.

First Embodiment

- (10) Prior to describing the features of the vehicular air conditioning system according to the present invention, the vehicular air conditioning system will be briefly described with reference to FIGS. 3 and 4.
- (11) The vehicular air conditioning system includes an intake part **10** that sucks an indoor air or an outdoor air and blows it into the vehicle interior.
- (12) The intake part **10** is installed on the engine room side with respect to a dash panel D. The intake part **10** includes an intake case **12**. The intake case **12** is provided with an indoor air inlet **12a** and an outdoor air inlet **12b**. An intake door **14** is installed between the indoor air inlet **12a** and the outdoor air inlet **12b**.
- (13) The indoor air inlet **12a** communicates with the vehicle interior to introduce an indoor air, and the outdoor air inlet **12b** communicates with the vehicle exterior to introduce an outdoor air.
- (14) The indoor air inlet **12a** is formed on the side surface portion of the intake case **12** corresponding to the vehicle interior, and the outdoor air inlet **12b** is formed on the upper surface portion of the intake case **12** corresponding to the vehicle exterior. Therefore, the outdoor air inlet **12b** is located more upward than the indoor air inlet **12a** in the gravity direction.
- (15) The intake door **14** is a dome-shaped door that rotates between the outdoor air inlet **12b** and the indoor air inlet **12a** to selectively open the outdoor air inlet **12b** or the indoor air inlet **12a**.
- (16) In particular, the intake door **14** blocks the outdoor air inlet **12b** as it moves toward the upper surface portion of the intake case **12**, and blocks the indoor air inlet **12a** as it moves toward the side surface portion of the intake case **12**.
- (17) In this regard, the intake door **14** includes side plate portions **16** arranged on both sides and a

dom-shaped upper plate portion **18** configured to connect the side plate portions **16**.

(18) The side plate portions **16** on both sides have rotation center shafts **16a** rotatably fitted into the shaft holes **12e** of the intake case **12**. The side plate portions **16** on both sides assembled in this way are rotated about the rotation center shafts **16a** to rotate the intake door **14**.

(19) The upper plate portion **18** directly blocks the outdoor air inlet **12b** or the indoor air inlet **12a**. In particular, the intake door **14** blocks the outdoor air inlet **12b** as it is erected upward, and blocks the indoor air inlet **12a** as it lays to the side.

(20) Meanwhile, a seal **14a** is provided around the edges of the side plate portions **16** and the upper plate portion **18**. The seal **14a** is in close contact with the inner wall of the intake case **12**. In particular, the seal **14a** is in close contact with the door sealing surface **12c** on the side of the outdoor air inlet **12b** side or the door sealing surface **12d** on the side of the outdoor air inlet **12b**, which are formed in the intake case **12**.

(21) The seal **14a** hermetically seals a gap between the intake door **14** and the outdoor air inlet **12b**, or a gap between the intake door **14** and the indoor air inlet **12a**.

(22) In addition, the intake part **10** includes an air filter **20** and a blower **22**. The air filter **20** installed on the downstream side of the intake door **14** to filter foreign substances in the indoor air or the outdoor air introduced through the indoor air inlet **12a** or the outdoor air inlet **12b**. The blower **22** sucks the indoor air or the outdoor air through the indoor air inlet **12a** or the outdoor air inlet **12b**, and blows the sucked indoor air or outdoor air toward the vehicle interior.

(23) Next, the features of the vehicular air conditioning system according to the present invention will be described in detail with reference to FIGS. **3** to **5**.

(24) First, the vehicular air conditioning system according to the present invention includes a water blocking part **30** capable of blocking water flowing into the vehicle interior from the indoor air inlet **12a** of the intake case **12**.

(25) The water blocking part **30** includes a water storage groove **32** formed on the bottom surface **12a-1** of the indoor air inlet **12a** of the intake case **12** facing the vehicle interior.

(26) The water storage groove **32** is formed along the width direction of the indoor air inlet **12a** at a predetermined depth on the water flow path between the outdoor air inlet **12b** and the indoor air inlet **12a** of the intake case **12** on the bottom surface **12a-1** of the indoor air inlet **12a** of the intake case **12**.

(27) In particular, the water introduced into the outdoor air inlet **12b** flows toward the bottom surface **12a-1** of the indoor air inlet **12a** through the upper surface of the intake door **14** and the door sealing surfaces **12c** and **12d** of the intake case **12** and then flows into the vehicle interior. The water storage groove **32** is formed on the water flow path extending from the outdoor air inlet **12b** to the indoor air inlet **12a**.

(28) Since the water storage groove **32** is formed on the bottom surface **12a-1** of the indoor air inlet **12a**, the water storage groove **32** serves to store the water **W** moved from the outdoor air inlet **12b** to the indoor air inlet **12a** through the upper surface of the intake door **14** and the door sealing surfaces **12c** and **12d** of the intake case **12**.

(29) In particular, when the water introduced into the outdoor air inlet **12b** is accumulated on the upper surface of the intake door **14**, the water **W** is poured onto the bottom surface **12a-1** of the indoor air inlet **12a** during the rotational movement of the intake door **14**. The water storage groove **32** stores the water **W** poured into the indoor air inlet **12a**.

(30) In addition, the water introduced into the outdoor air inlet **12b** flows into the vehicle interior while moving toward the indoor air inlet **12a** through the door sealing surfaces **12c** and **12d** of the intake case **12**. The water storage groove **32** stores the water **W** moving from the door sealing surfaces **12c** and **12d** to the indoor air inlet **12a**.

(31) Therefore, it is possible to prevent the movement of the water from the outdoor air inlet **12b** to the indoor air inlet **12a**, thereby preventing the water from flowing from the outdoor air inlet **12b** into the vehicle interior.

(32) Referring again to FIGS. 3 to 5, the water blocking part **30** further includes a drain passage **40** capable of draining the water **W** collected in the water storage groove **32**.

(33) The drain passage **40** includes a drain hole **42** formed on the bottom surface **32a** of the water storage groove **32**. The drain hole **42** is formed to extend from the bottom surface **32a** of the water storage groove **32** toward the inner lower side of the intake case **12**. The drain hole **42** drains the water **W** collected in the water storage groove **32** into the intake case **12**. In particular, the water **W** is drained downward along the inner wall surface of the intake case **12**.

(34) Accordingly, the water **W** drained to the lower portion of the intake case **12** is allowed to flow into the inner passage **12f** of the intake case **12**, and can move to an evaporator condensate drain hole (not shown) on the downstream side of the inner passage **12f**. As a result, the water **W** can be discharged to the outside through the evaporator condensate drain hole.

(35) Meanwhile, the drain hole **42** is formed on the bottom surface **32a** of the water storage groove **32**, and is preferably formed in a part of the bottom surface **32a** of the water storage groove **32**. In some cases, a plurality of drain holes **42** may be formed at intervals on the bottom surface **32a** of the water storage groove **32**.

(36) In this regard, the bottom surface **32a** of the water storage groove **32** is inclined downward toward the drain hole **42**. Therefore, the water **W** stored in the water storage groove **32** can be quickly drained to the drain hole **42**.

(37) In addition, the drain hole **42** is preferably formed to extend from the bottom surface **32a** of the water storage groove **32** toward the inner lower side of the intake case **12** so that the end portion **42a** of the drain hole **42** faces the lateral edge of the air filter **20** on the downstream side thereof.

(38) More preferably, the end portion **42a** of the drain hole **42** is formed to face the edge band **20a** of the air filter **20**. In particular, the end portion **42a** of the drain hole **42** extends toward the underside of the band **20a** of the air filter **20** so as to be covered by the band **20a** of the air filter **20**.

(39) The reason for adopting this configuration is to ensure that the water **W** drained from the water storage groove **32** does not flow into the inner filter portion **20b** of the air filter **20**, and does not obstruct the flow of an air sucked into the inner filter portion **20b** of the air filter **20**.

(40) In particular, the water **W** drained from the water storage groove **32** may be scattered by the wind pressure of the air sucked into the inner filter portion **20b** of the air filter **20**. The above configuration can prevent such scattering of the water.

(41) Referring again to FIGS. 3 to 5, the water blocking part **30** further includes a water blocking wall **50** formed in the intake case **12** between the water storage groove **32** and the indoor air inlet **12a**.

(42) The water blocking wall **50** protrudes from the bottom surface **12a-1** on the side of the indoor air inlet **12a** to the indoor air inlet **12a**, and is formed long along the width direction of the indoor air inlet **12a**.

(43) As shown in FIGS. 3 and 5, the water blocking wall **50** serves to secondarily block the water **W** introduced from the outdoor air inlet **12b** toward the indoor air inlet **12a** through the water storage groove **32**.

(44) Therefore, when excessive water **W** is introduced from the outdoor air inlet **12b** or when the motor vehicle is excessively tilted downward toward the rear side, the water blocking wall **50** secondarily blocks the water flowing into the indoor air inlet **12a** through the water storage groove **32**. As a result, it is possible to reliably prevent the water **W** from flowing from the indoor air inlet **12a** toward the vehicle interior.

(45) Meanwhile, the water blocking wall **50** protrudes from the bottom surface **12a-1** on the side of the indoor air inlet **12a** toward the indoor air inlet **12a** and has a structure and height so as not to impede the flow of an air introduced from the vehicle interior to the indoor air inlet **12a**.

(46) According to the air conditioner of the first embodiment having such a configuration, the water **W** moving from the outdoor air inlet **12b** toward the indoor air inlet **12a** is stored and then quickly

drained. Therefore, it is possible to limit the movement of the water W from the outdoor air inlet **12b** toward the indoor air inlet **12a**.

(47) In addition, since the movement of water from the outdoor air inlet **12b** to the indoor air inlet **12a** can be limited, it is possible to reliably prevent the flow of water W from the outdoor air inlet **12b** into the indoor air inlet **12a** and consequently the flow of water W from the outdoor air inlet **12b** into the vehicle interior.

Second Embodiment

(48) Next, a vehicular air conditioning system according to a second embodiment of the present invention will be described with reference to FIGS. **6** to **8**.

(49) The vehicular air conditioning system according to the second embodiment includes a drain part **60** capable of draining water introduced onto the upper surface of the intake door **14** to the lower side of the intake case **12**.

(50) The drain part **60** includes first drain passages **62** for draining the water flowing onto the upper plate portion **18** of the intake door **14** toward the side plate portions **16**, second drain passages **64** for draining the water drained toward the rotation center shafts **16a** of the side plate portions **16**, and third drain passages **66** for draining the water drained to the rotation center shafts **16a** toward the inner wall portions of the intake case **12**.

(51) The first drain passages **62** include a plurality of water collection grooves **62a** for collecting the water introduced into the dome-shaped upper plate portion **18** of the intake door **14** toward both ends of the intake door **14** in the rotational direction, and drain grooves for draining the water collected at both ends of the upper plate portion **18** toward the left and right side plate portions **16**, respectively.

(52) The water collection grooves **62a** are formed to be inclined downward at a predetermined angle from the central portion of the upper plate portion **18** in the rotational direction toward the front and rear ends of the upper plate portion **18** in the rotational direction. The water collection grooves **62a** formed in this way collect the water introduced onto the upper plate portion **18** to the front and rear ends of the upper plate portion **18**, respectively.

(53) The drain grooves **62b** are formed at the front and rear ends of the upper plate portion **18** in the rotational direction, respectively, and are formed to be inclined downward at a predetermined angle from the central portion of the upper plate portion **18** in the left-right direction toward the side plate portions **16**. The drain grooves **62b** formed in this way drain the water collected at the front and rear ends of the upper plate portion **18** toward the side plate portions **16**, respectively.

(54) The second drain passages **64** are formed to extend from the drain grooves **62b** of the upper plate portion **18** toward the rotation center shafts **16a** of the side plate portions **16**. The second drain passages **64** thus formed drain the water on the side of the drain grooves **62b** toward the rotation center shafts **16a** of the side plate portions **16**.

(55) The third drain passages **66** are formed on the inner wall portions of the intake case **12** corresponding to the rotation center shafts **16a** of the side plate portions **16**. In particular, the third drain passages **66** are formed on the inner wall portions of the intake case **12** corresponding to the rotation center shafts **16a** corresponding to the lower portions of the rotation center shafts **16a**. The end portions **66a** of the third drain passages **66** are formed to face the lower side of the intake case **12**.

(56) The third drain passages **66** drain the water W in the rotation center shafts **16a** of the intake door **14** along the inner wall surfaces of the intake case **12**. In particular, the third drain passages **66** drain the water W to the lower portion of the intake case **12** along the inner wall surfaces of the intake case **12**.

(57) Accordingly, the water W drained to the lower portion of the intake case **12** can flow into the inner passage **12f** of the intake case **12**, and can move to the evaporator condensate drain hole (not shown) on the downstream side of the inner passage **12f**. As a result, the water W can be discharged to the outside through the evaporator condensate drain hole.

(58) Meanwhile, it is preferable that the third drain passages **66** are formed downward from the inner wall surfaces of the intake case **12** on the side of the rotation center shafts **16a** of the intake door **14**, and the end portions **66a** of the third drain passages **66** are formed to face the side edges of the air filter **20** on the downstream side thereof.

(59) More preferably, the end portions **66a** of the third drain passages **66** are formed to face the edge band **20a** of the air filter **20**. In particular, the end portions **66a** of the third drain passages **66** extend toward the underside of the bands **20a** of the air filter **20** so as to be covered by the bands **20a** of the air filter **20**.

(60) The reason for adopting this configuration is to ensure that the water **W** drained to the rotation center shafts **16a** of the intake door **14** does not flow into the inner filter portion **20b** of the air filter **20** and does not disturb the flow of an air sucked into the inner filter portion **20b** of the air filter **20**.

(61) In particular, the water drained to the rotation center shafts **16a** of the intake door **14** may be scattered by the wind pressure of the air sucked into the inner filter portion **20b** of the air filter **20**. The above configuration can prevent the scattering of water.

(62) In addition, the third drain passages **66** are formed downward on the inner wall surfaces of the intake case **12** on the side of the rotation center shafts **16a** of the intake door **14**. As shown in FIGS. **3** and **5**, the third drain passages **66** are formed between the door sealing surface **12c** on the side of the outdoor air inlet **12b** and the door sealing surface **12d** on the side of the indoor air inlet **12a**, which are formed in the intake case **12**.

(63) In particular, the door sealing surface **12c** on the side of the outdoor air inlet **12b** and the door sealing surface **12d** on the side of the indoor air inlet **12a** of the intake case **12** are formed to correspond to each other on the inner wall surfaces of the intake case **12** with respect to the rotation center shafts **16a** of the intake door **14**. At this time, the third drain passages **66** are formed to isolate the door sealing surface **12c** on the side of the outdoor air inlet **12b** and the sealing surface **12d** on the side of the indoor air inlet **12a** from each other.

(64) The reason for adopting this configuration is to prevent the water introduced onto the upper surface of the intake door **14** from flowing toward the door sealing surface **12d** on the side of the indoor air inlet **12a** while moving along the door sealing surface **12c** on the side of the outdoor air inlet **12b**.

(65) Therefore, it is possible to prevent the water introduced onto the upper surface of the intake door **14** from flowing into the vehicle interior along the door sealing surface **12c** on the side of the outdoor air inlet **12b** and the sealing surface **12d** on the side of the indoor air inlet **12a** of the intake case **12**.

(66) According to the vehicular air conditioning system of the present invention having such a configuration, the water introduced onto the upper surface of the intake door **14** can be guided toward both inner wall surfaces of the intake case **12** and can be quickly drained downward. Therefore, it is possible to prevent water from being accumulated on the upper surface of the intake door **14**.

(67) In addition, since the accumulation of water on the upper surface of the intake door **14** can be prevented, it is possible to prevent the water accumulated on the upper surface of the intake door **14** from flowing into the vehicle interior.

(68) While the preferred embodiments of the present invention have been described above, the present invention is not limited to the above-described embodiments. Various modifications and changes may be made without departing from the scope and spirit of the present invention defined in the claims.

(69) For example, it is shown in the drawings that the water blocking part **30** of the first embodiment and the drain part **60** of the second embodiment are independently installed in the intake part **10** of the vehicular air conditioning system. However, the present invention is not limited thereto. The water blocking part **30** of the first embodiment and the drain part **60** of the

second embodiment may be simultaneously installed in the intake part **10** of the vehicular air conditioning system.

Claims

1. A vehicular air conditioning system, comprising: an intake case having an indoor air inlet for introducing indoor air present in a vehicle interior and an outdoor air inlet for introducing outdoor air present outside a vehicle; an intake door installed in the intake case to selectively block one of the outdoor air inlet and the indoor air inlet while rotating about a rotation center shaft; and a water blocking part configured to block water flowing into the vehicle interior from the indoor air inlet of the intake case, wherein the outdoor air inlet is located more upward than the indoor air inlet in a gravity direction, and wherein the water blocking part includes a water storage groove formed in a portion of the intake case below the indoor air inlet in a water passage between the outdoor air inlet and the indoor air inlet, and configured to store the water moving from the outdoor air inlet toward the indoor air inlet.
2. The system of claim 1, wherein the water storage groove is configured to store the water that flows toward a bottom surface of the indoor air inlet when the water accumulated on the upper surface of the intake door flows to the bottom surface of the indoor air inlet during the rotational movement of the intake door.
3. The system of claim 2, wherein an outdoor air inlet side door sealing surface and an indoor air inlet side door sealing surface on which the intake door is seated and sealed are formed on an inner wall surface of the intake case, and the water storage groove is configured to store the water that flows from the outdoor air inlet toward the indoor air inlet along the outdoor air inlet side door sealing surface and the indoor air inlet side door sealing surface of the intake case.
4. The system of claim 3, further comprising: a drain passage configured to drain the water collected in the water storage groove.
5. The system of claim 4, wherein the drain passage includes a drain hole formed to extend from a bottom surface of the water storage groove toward an inner lower side of the intake case to drain the water stored in the water storage groove to the inner lower side of the intake case.
6. The system of claim 5, wherein an air filter is installed on the downstream side of the intake door, and the drain hole has an end portion formed to face a side edge of the air filter on the downstream side thereof.
7. The system of claim 6, wherein the end portion of the drain hole extends toward a lower side of a band of the air filter so as to be covered by the band of the air filter.
8. The system of claim 3, wherein the water storage groove has a bottom surface inclined downward toward a drain hole.
9. The system of claim 3, further comprising: a water blocking wall formed in a portion of the intake case between the water storage groove and the indoor air inlet, wherein the water blocking wall protrudes from a bottom surface below the indoor air inlet toward the indoor air inlet to secondarily block the water flowing toward the indoor air inlet through the water storage groove.
10. A vehicular air conditioning system, comprising: an intake case having an indoor air inlet for introducing indoor air present in a vehicle interior and an outdoor air inlet for introducing outdoor air present outside a vehicle; an intake door installed in the intake case to selectively block one of the outdoor air inlet and the indoor air inlet while rotating about a rotation center shaft; a water blocking part configured to block water flowing into the vehicle interior from the indoor air inlet of the intake case; and a drain part configured to guide the water introduced onto an upper surface of the intake door through the outdoor air inlet toward an inner wall of the intake case and drain the water downward, wherein the outdoor air inlet is located more upward than the indoor air inlet in a gravity direction, and wherein the intake door includes side plate portions configured to rotate about the rotation center shaft between the outdoor air inlet and the indoor air inlet, and a dome-

shaped upper plate portion configured to block the outdoor air inlet when the side plate portions move toward the outdoor air inlet, and configured to block the indoor air inlet when the side plate portions move toward the indoor air inlet.

11. The system of claim 10, wherein the drain part includes first drain passages for draining the water flowing onto the upper plate portion of the intake door toward the side plate portions, second drain passages for draining the water from the side plate portions toward the rotation center shaft of the side plate portions, and third drain passages for draining the water from the side plate portions toward the inner wall of the intake case.

12. The system of claim 11, wherein the first drain passages include a plurality of water collection grooves for collecting the water introduced into the upper plate portion of the intake door toward front and rear ends of the intake door in the rotational direction, and drain grooves for draining the water collected at the front and rear ends of the upper plate portion toward the side plate portions.

13. The system of claim 12, wherein the plurality of water collection grooves are formed to be inclined downward at a predetermined angle from a central portion of the upper plate portion in the rotational direction toward the front and rear ends of the upper plate portion in the rotational direction and configured to collect the water introduced onto the upper plate portion to the front and rear ends of the upper plate portion, and the drain grooves are formed at the front and rear ends of the upper plate portion in the rotational direction and formed to be inclined downward at a predetermined angle from the central portion of the upper plate portion in the left-right direction toward the side plate portions to drain the water collected at the front and rear ends of the upper plate portion toward the side plate portions.

14. The system of claim 13, wherein the second drain passages are formed to extend from the drain grooves of the upper plate portion toward the rotation center shafts of the side plate portions and configured to drain the water on side surfaces of the drain grooves toward the rotation center shafts of the side plate portions.

15. The system of claim 14, wherein the third drain passages are formed on the inner wall portions of the intake case corresponding to the rotation center shafts of the side plate portions to drain the water from the rotation center shafts of the side plate portions to the inner wall surfaces of the intake case.

16. The system of claim 15, wherein the third drain passages are formed on the inner wall portions of the intake case corresponding to lower portions of the rotation center shafts, and the third drain passages have end portions formed to face a lower side of the intake case to drain the water downward from the rotation center shafts of the intake door along the inner wall surfaces of the intake case.

17. The system of claim 16, wherein an air filter is installed on the downstream side of the intake door, and the end portions of the third drain passages are formed to face a side edge of the air filter.

18. The system of claim 17, wherein the end portions of the third drain passages extend to below a band of the air filter so as to be covered by the band of the air filter.

19. The system of claim 11, wherein an outdoor air inlet side door sealing surface and an indoor air inlet side door sealing surface are formed on the inner wall surfaces of the intake case, the intake door is configured to block the outdoor air inlet or the indoor air inlet while being seated on the outdoor air inlet side door sealing surface or the indoor air inlet side door sealing surface of the intake case, and the third drain passages are formed on inner wall surfaces of the intake case corresponding to the rotation center shafts of the intake door so as to isolate the outdoor air inlet side door sealing surface and the indoor air inlet side door sealing surface from each other to prevent movement of the water from the outdoor air inlet side door sealing surface toward the indoor air inlet side door sealing surface.
